INTRODUCTION TO INFORMATION GEOMETRY

Gaussian Geodesics and Statistical Distance

Week 10 · Lecture 2

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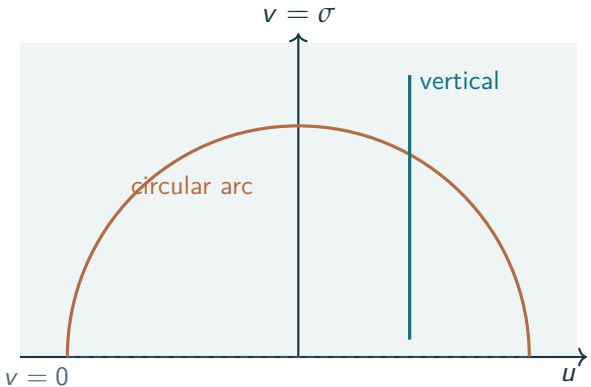
Geodesics in the upper half-plane

In coordinates $u = \mu/\sqrt{2}$, $v = \sigma$,

$$ds_F^2 = 2\frac{du^2 + dv^2}{v^2}.$$

Geodesics are vertical lines and arcs of circles whose centers lie on $v = 0$. The circular arcs meet the boundary at right angles.

A geometric picture of Gaussian geodesics



Why a curved route can win

At height v, a small horizontal step du costs

$$ds_F = \sqrt{2}\frac{|du|}{v}.$$

A path can trade some vertical travel for cheaper horizontal travel by passing through larger standard deviations. The semicircular geodesic balances these contributions.

Vertical geodesics and scale changes

For two Gaussians with the same mean, the shortest route is vertical in the half-plane. Its length is

$$d_F\big((\mu,\sigma_1),(\mu,\sigma_2)\big) = \sqrt{2}\left|\ln\frac{\sigma_2}{\sigma_1}\right|.$$

Distance between two centered Gaussians

Find the Fisher distance between

$$N(0,1) \qquad \text{and} \qquad N(0,4).$$

Use the standard deviations in

$$d_F = \sqrt{2}\,|\ln(\sigma_2/\sigma_1)|\,.$$

Distance between two centered Gaussians

The two standard deviations are $\sigma_1 = 1$ and $\sigma_2 = 2$. Therefore

$$d_F = \sqrt{2} \ln 2 \approx .9803.$$

Doubling standard deviation requires the same Fisher length, regardless of the starting scale.

Constant Fisher speed on a vertical route

A suitable time parameter is

$$v(t) = v_1 \exp\left(t \ln \frac{v_2}{v_1}\right), \qquad 0 \le t \le 1.$$

Indeed $v'(t)/v(t) = \ln(v_2/v_1)$, so

$$\sqrt{2}\frac{|v'(t)|}{v(t)} = \sqrt{2}\left|\ln \frac{v_2}{v_1}\right|.$$

A constant-speed scale midpoint

Let $\sigma(0) = 1$, $\sigma(1) = 4$, with fixed mean. Find the midpoint $\sigma(1/2)$ along the constant-Fisher-speed geodesic. How does it compare with the arithmetic midpoint 2.5?

A constant-speed scale midpoint

With $\sigma(t) = \exp(t \ln 4) = 4^t$,

$$\sigma(1/2) = \sqrt{4} = 2.$$

The Fisher midpoint in scale is the geometric mean $\sqrt{\sigma_1 \sigma_2}$, rather than the arithmetic mean.

A formula for any pair of Gaussians

Write $P_i = (\mu_i, \sigma_i)$, with $\sigma_i > 0$. Their Fisher distance is

$$d_F(P_1, P_2) = \sqrt{2}\ \operatorname{arcosh}\left(1 + \frac{(\mu_1 - \mu_2)^2 + 2(\sigma_1 - \sigma_2)^2}{4\sigma_1\sigma_2}\right).$$

The expression inside arcosh is at least one.

Equal-width Gaussians, separated means

Take two distributions with

$$(\mu_1, \sigma_1) = (-\sqrt{2}, 1), \quad (\mu_2, \sigma_2) = (\sqrt{2}, 1).$$

Evaluate the general Gaussian Fisher distance formula.

Equal-width Gaussians, separated means

Here $(\mu_1 - \mu_2)^2 = 8$ and $\sigma_1 = \sigma_2 = 1$. The arcosh argument is $1 + 8/4 = 3$, so

$$d_F = \sqrt{2}\,\operatorname{arcosh}(3) \approx 2.493.$$

The geodesic between those two laws

With $u = \mu/\sqrt{2}$, the endpoints are $(-1, 1)$ and $(1, 1)$. The geodesic lies on the semicircle

$$u^2 + v^2 = 2, \qquad v > 0.$$

At its highest point, $v = \sqrt{2}$; the geodesic temporarily uses a wider Gaussian than either endpoint.

Relative entropy between two Gaussians

For $P = N(\mu_1, \sigma_1^2)$ and $Q = N(\mu_2, \sigma_2^2)$,

$$D(P\|Q) = \ln \frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}.$$

Unlike the Fisher distance, KL divergence can change when the two distributions are interchanged.

Compare both KL directions

Let $P = N(0, 1)$ and $Q = N(0, 4)$. Using the Gaussian KL formula, find

$$D(P\|Q) \quad \text{and} \quad D(Q\|P).$$

Which is larger?

Compare both KL directions

For $P = N(0, 1)$ and $Q = N(0, 4)$,

$$D(P\|Q) = \ln 2 - \frac{3}{8} \approx .318,$$

$$D(Q\|P) = \frac{3}{2} - \ln 2 \approx .807.$$

The wider law is further from the narrow one in this KL direction.

Two different kinds of boundary

In the Gaussian family, the limit $\sigma \to 0$ is at infinite Fisher distance: $\sqrt{2}|\ln \sigma| \to \infty$. In a finite categorical family, a probability coordinate can approach zero in finite Fisher distance. Both spaces carry Fisher metrics, yet their global geometries differ.

A geometric map of the course

1. Replicator equations describe motion of distributions.
2. Entropy and KL divergence measure information and discrepancy.
3. The Fisher metric measures changes in probability laws.
4. Natural gradients and geodesics use that metric to describe motion and distance.

One last comparison

The categorical Fisher manifold is a spherical orthant; the Gaussian location-scale family is a hyperbolic half-plane. On both spaces, a point represents a distribution, a tangent vector represents its change, and a metric measures that change. That common structure is the geometry of information.